MIL-C-23771A(AS)
10 August 1973
SUPERSEDING
MIL-C-23771 (Wep)
24 June 1963

MILITARY SPECIFICATION

CONTROL UNIT

(FOR MK-7 RETRACTABLE DECK SHEAVE)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes requirements for a control unit used with the MK-7 retractable deck sheave.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

FF-B-171	Bearings, Ball, Annular (General Purpose)
FF-B-185	Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning
NN-P-530	Plywood, Flat Panel
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated, Plywood

FSC 1710

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SPECIFICATIONS

Federal (Contd)

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-C-843	Cushioning Material, Cellulosic

Military

MIL-P-116	Preservation, Methods of
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-B-233	Boxes, Repair Parts, Storage
MIL-S-901	Shock Tests, H. I. (High Impact); Shipboard Machinery Equipment and Systems, Requirements for
MIL-D-1000/1	Drawing, Engineering and Associated Lists
MIL-D-1000/2	Drawing, Engineering and Associated Lists
MIL-C-2212	Controller, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity
MIL-M-8910	Manuals, Technical, Illustrated Parts Breakdown
MIL-L-15016	Lubricating Oil, General Purpose
MIL-P-15024	Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment

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SPECIFICATIONS

Military (Contd)

MIL-E-15090	Enamel, Equipment, Light Gray (Formula No. 11)
MIL-L-15719	Lubricating-Grease (High-Temperature, Electric Motor, Ball and Roller Bearings)
MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback Cold-Application
MIL-M-17060	Motors, 60 cycle, Alternating Current, Integral HP (Shipboard Use)
MIL-S-19622	Stuffing Tubes, Nylon
MIL-G-81322	Grease, Aircraft, General Purpose, Wide Temperature Range

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods

(When requesting applicable documents, refer to both title and number. Copies of this specification and applicable specifications and standards may be obtained upon application to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.)

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DRAWINGS

Naval Air Engineering Center, Engineering Department (Ship Installations)

71-40891 Control Unit, Retractable Deck Sheave

(Copies of the applicable NAEC drawings may be obtained upon application to Commanding Officer, Naval Air Engineering Center, Engineering Department (SI), Philadelphia, Pennsylvania 19112; Attention: Technical Data Branch.)

3. REQUIREMENTS

3.1 General - The control unit shall consist of an electrically driven operating mechanism complete with limit and torque switches, and means for manual operation.

3.2 Materials - Materials shall conform to applicable specifications as specified herein and on the applicable drawings. Materials which are not covered by applicable specifications or drawings or which are not specifically described herein shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended. The material shall be free from all defects and imperfections that may affect the quality and reliability of the finished product.

3.2.1 Castings - Castings shall be of high quality, smooth, sound, and free from blow holes, porosity, cracks, or other defects.

3.3 Selection of specifications and standards - Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Design and construction - The control unit is an electrical reversible power drive complete with an auxiliary manual handwheel control and shall consist of motor unit, push button station, and controller. The drive shall be electrically operated in either direction from commands initiated at the push button station through electrical latching-in of selected magnetic contactor in the controller and automatically operated until either the torque switch opens in the extreme "down" position or limit switch actuates in the extreme "up" position as shown in the schematic diagram of NAEC Drawing 71-40891. The motor drive shall also be stopped by either manual operation of the "STOP" control at the push button station or automatic operation of the normally closed overload contacts of the heater elements.

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3.4.1 Motor unit - The motor unit shall conform to the outline dimensional requirements and specification of NAEC Part No. 71-40891-4. It shall consist of a motor drive unit, reduction gearing and housing, drive sleeve, torque switch, and limit switch.

3.4.1.1 Driving motor - The driving motor shall be of sufficient capacity to satisfactorily operate the unit but shall not exceed 5 HP rated output. Nominal stall torque to be 25 Ft-Lbs. The motor shall conform to all requirements of MIL-M-17060 and to the following: Service "A", 40° C ambient temperature, 440 volt, short time (15 minutes) duty, totally enclosed, 1800 synchronous RPM, constant speed, 3-phase squirrel cage induction, ball bearing double sealed single-row, Class B insulation, reversible, high-torque, normal starting current, horizontal, flange mounting.

3.4.1.2 Reduction gearing and housing - The reduction gearing shall consist of worms and/or gears as necessary to provide the performance specified. The entire gear train shall be housed in a water-tight housing. Provision shall be made within the housing for adequate lubrication of the gear system with lubricating oil conforming to MIL-L-15016 or grease conforming to MIL-G-81322.

3.4.1.2.1 Housing mounting - The housing shall further provide for the mounting of the driving motor, limit switch, torque switch, manual control, and for flange mounting of the unit. It shall have dimensions as specified on NAEC Drawing 71-40891.

3.4.1.3 Drive sleeve - The drive sleeve shall conform to the strength requirements of this specification and shall have a bore and key seat in accordance with dimensions shown for NAEC Part No. 71-40891-4.

3.4.1.4 Switches - A torque switch and a limit switch shall be provided to stop the driving motor. Their operation shall conform to the performance requirements of this specification. Switches shall be contained within a waterproof housing mounted on the gear case, having a removable cover for access to switch adjustments, and for the use of standard Navy waterproof stuffing tubes conforming to MIL-S-19622. Direction of rotation of the drive sleeve to actuate the limit switch shall be opposite to the direction to actuate the torque switch.

3.4.1.4.1 Torque switch - An adjustable torque sensing switch shall be provided capable of safely interrupting the control circuit and stopping the motor in the direction specified in NAEC Drawing 71-40891. The adjustable range shall be between a minimum of 40 foot-pounds and a maximum of 400 foot-pounds, plus or minus 10 percent, measured at the drive sleeve. The range selector shall have

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a numerically graduated scale and a position indicator. The torque switch shall open its normally closed contacts and stop the driving motor when the pre-set drive sleeve torque load setting has been attained. The torque switch drive shall be either gear or roller actuated to minimize wear by repeated actuations. The mechanism and gear drive which actuates the torque switch shall be non-self-reversing so that the contacts of the torque switch will be maintained open until the opposite raise command.

3.4.1.4.2 Limit switch - A limit switch shall be provided consisting of two adjustable sections. Each section shall consist of a minimum of three sets of contacts (one normally open and two normally closed) which when actuated at the other extreme end position produces two open and one closed set of contacts for that section. One section of limit switch shall operate in one extreme clockwise shaft position and the other section of limit switch shall operate in the other extreme counterclockwise shaft position. The limit switch contacts shall be capable of safely interrupting the control circuit and stopping the motor when the drive is in the extreme clockwise position of the output drive shaft after the torque switch is closed and the drive sleeve makes 9-3/4 ± 1/4 revolutions.

3.4.1.5 Manual operation - A hand wheel shall be provided to rotate the drive sleeve in either direction in case of power failure. Construction shall be such that the hand wheel does not revolve during normal motor operation, and the motor shall be mechanically disengaged during manual operation. If the hand wheel is engaged, it shall automatically be disengaged when the motor is operated. Tools should not be required in changing to or from manual operation. A maximum force of 25 pounds applied at the hand wheel shall exert a minimum torque of 75 foot-pounds at the drive sleeve. The hand wheel shall be readily removable and shall be capable of being rapidly reinstalled and engaged as an integral unit. No loose method of attachment such as keys or cotter pins shall be utilized.

3.4.1.6 Output - The motorized unit shall be capable of delivering a minimum of 70 foot-pounds of torque at a constant full load speed of 60 RPM minimum at the driving sleeve. The minimum starting torque shall be 350 foot-pounds.

3.4.1.7 Shock test - The motorized unit shall pass the shock test requirements of MIL-S-901 as outlined in 4.5.4. This requirement may be waived for units that have previously passed this shock test providing that subsequently its design has not been significantly altered.

3.4.2 Controller (NAEC Part No. 71-40891-3) - A controller shall be supplied of sufficient capacity to properly handle operation of the driving motor. The controller shall conform to all requirements of MIL-C-2212 and to the following: Size number 0, magnetic operation. across-line type, semi-automatic performance,

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motor starting and reversing controller function. 440 volts, 3-phase, intermittent duty, low-voltage protection, pilot-circuit protection, emergency run feature, drip-proof enclosure. The controller shall be arranged for use with a three position (RAISE-STOP-LOWER) push button station. The controller shall provide for proper control of the motor as required by the torque and limit switches. The controller shall serve to operate the motor when either the RAISE or LOWER pushbutton is depressed and shall keep the motor in operation until torque, limit, or STOP pushbutton is operated or overload relay is opened. Controller shall be suitable for operation described and arrangement shown on NAEC Drawing 71-40891.

3.4.3 Pushbutton station (NAEC Part No. 71-40891-2) - A pushbutton station shall be supplied of sufficient capacity to properly handle operation of the driving motor by means of the controller. The pushbutton station shall conform to requirements of MIL-C-2212 and to the following: 440 volts single-phase, remote location, momentary contact pushbutton type, intermittent duty, watertight, manual operation, surface mounting. Station shall contain three elements; two elements (RAISE-LOWER) shall have two sets of double break normally open contacts, one element (STOP) shall have one set of double break normally closed contacts. Buttons shall be identified as indicated by NAEC Part No. 71-40891-2.

3.4.4 Plans - The manufacturer shall furnish with his quotation sufficient information in duplicate to enable the Government to obtain a clear understanding of the equipment he proposes to furnish. This information may be in the form of blueprint plans or typewritten or printed specifications, in such arrangement as the manufacturer may prefer. Quotations not completely indicating the character of the proposed equipment may be rejected as incomplete. This information shall consist of the following and shall include any other data deemed necessary for a complete determination of the construction and operating characteristics.

(a) Outline, overall and principal dimensions
(b) Weights
(c) Complete rating characteristics
(d) Complete description of operation including wiring diagram
(e) An itemized list of spare parts recommended by the bidder for shipboard maintenance and replacement which he will furnish if requested
(f) A definite statement as to whether or not any exceptions are being taken to applicable specifications or drawings with full description of such exceptions and consequent departure from the specifications or drawings, if any.

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3.4.4.1 Approval plans - When required the contractor shall submit for the approval of the Naval Air Engineering Center, Engineering Department (SI), (Code NE-7) Philadelphia, Pennsylvania 19112, via the Navy Inspector, plans of the equipment involved as soon as possible after the contract award. The manufacturer shall not proceed with the manufacture or shipment of such equipment prior to approval of these plans by the Government, unless otherwise authorized. Approval of these plans shall in no way relieve the contractor of responsibility of conformance to this specification. Plans for the motor, controller and pushbutton station shall be as specified in the specifications previously referenced in items 3.4.1.1, 3.4.2 and 3.4.3. The plan for the motor unit shall provide an outline of the assembly including space envelope for removal of motor. In addition, an assembly of the reduction gear and housing with cross sectional views shall be shown. All component parts shall be called out and referenced to a bill of material. The bill of material shall also reference manufacturer's part number designation and military equivalent where known.

3.4.4.2 Record plans -

(a) Certification data sheet - Certification data sheets shall be furnished for the motor unit, driving motor, pushbutton station and controller. Where the detail specification does not outline format and contents of certification data sheets, the requirements of MIL-D-1000/2 shall apply.

(b) Master plans - When required reproducible copies of drawings approved in paragraph 3.4.4.1 above shall be furnished for record.

3.5 Spare parts - Spare parts shall be furnished only when required by the procuring activity.

3.5.1 Electrical spares - Electrical spare parts when required shall be furnished in accordance with MIL-M-17060 and MIL-C-2212. (See 6.2.)

3.5.2 Spares boxing and tools - All spare parts shall be in a single box in accordance with MIL-B-233 and plainly stenciled "Arresting Gear Spare Parts - Control Unit (for MK-7 Retractable Deck Sheave)." All tools shall be contained in a suitable tool kit. All spare parts and tools shall be properly identified.

3.6 Wiring - All wiring integral with the unit or the unit components shall be furnished and installed by the contractor.

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3.7 Nameplates - Nameplates shall conform to the requirements of MIL-P-15024. A suitable nameplate shall be attached to the completed unit, in a readily accessible location, marked with this specification number, the manufacturer's name, identification symbols, serial number, Navy contract number, date of manufacture, NAEC Part No., Federal Stock No. and provide a blank space for the Inspector's official stamping.

3.8 Bearings - Ball and roller bearings shall conform to FF-B-171 and FF-B-185, respectively, and shall be used whenever practicable. Lubrication of bearings shall be in accordance with the requirements of MIL-L-15719.

3.9 Finish - After fabrication, the metallic surface of the control unit shall be in a clean and smooth condition and shall be painted with one coat of zinc chromate primer conforming to MIL-P-8585. After the primer coat has thoroughly dried one finish coat of light gray enamel, conforming to MIL-E-15090, shall be applied.

3.10 Interchangeability - All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the identification requirements of MIL-D-1000/1.

3.11 Performance characteristics - The entire control unit shall perform satisfactorily when subjected to the tests specified in Section 4.

3.12 Workmanship - The control unit including all parts and accessories, shall be fabricated, constructed and assembled to produce an item free from all defects which would affect proper functioning in service. Particular attention shall be given to neatness and thoroughness of welding, wiring, marking of parts and assemblies, finish, alignment of parts, tightness of screw assemblies, and freedom of parts from burrs and sharp edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspections - The inspection of control units shall be classified as follows:

(a) First article inspection - First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of contract to determine that the production item meets the requirements of this specification.

(b) Quality conformance inspection - Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lots with the requirements set forth in this specification.

4.3 First article inspection - First article inspection shall consist of the examinations and tests as specified in Table I. The inspections shall be made in the sequence shown in Table I.

TABLE I. First Article Inspection

Inspections	Reference Paragraph
Examination	4.5.1
Dimensional	4.5.2
Shock	4.5.4
Full load	4.5.3.1
Torque switch setting	4.5.3.2
Handwheel force	4.5.3.3
Smoothness of operation	4.5.3.4

4.3.1 First article sample - Unless otherwise specified, as soon as practicable after award of a contract or order, the contractor shall furnish a first article sample for inspection to determine conformance with this specification. The sample shall be representative of the construction, workmanship, components, and materials to be used during production. The sample shall consist of one complete control unit on contract or order. When a contractor is in continuous production of these control units from contract to contract, submission of further first article samples on the new contract may be waived at the discretion of the procuring activity (see 6.2). Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements of submitting to the quality conformance inspection. The sample shall be identified with the manufacturer's part number and contract number and shall be inspected as specified herein and in accordance with the contract or order (see 6.2).

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4.3.1.1 First article data and disposition - Upon completion of the first article inspection, all the applicable inspection reports and when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the Government quality control representative (see 6.3). The sample used for first article inspection shall not be considered as part of the quantity to be delivered under contract.

4.4 Quality conformance inspection - The examinations and tests comprising quality conformance inspection are classified as specified in Table II. The sampling and inspection levels shall conform to MIL-STD-105.

TABLE II. Quality Conformance Inspections

Inspections	Method	Sample Size	Acceptance Criteria 1/
Examination	4.5.1	Every control unit	Reject all defective units
Dimensional	4.5.2	Inspection Level S-3	Acceptable quality level 1.0
Full load	4.5.3.1	Inspection Level S-3	Acceptable quality level 1.0
Torque switch setting	4.5.3.2	Inspection Level S-3	Acceptable quality level 1.0
Handwheel force	4.5.3.3	Inspection Level S-3	Acceptable quality level 1.0
Smoothness of operation	4.5.3.4	Inspection Level S-3	Acceptable quality level 1.0

1/ The sampling plan acceptance numbers shall apply collectively to all the characteristics within a stated acceptable quality level.

4.5 Inspection methods -

4.5.1 Examination - Each assembled control unit shall be carefully examined to determine conformance to all the requirements of this specification for which no specific tests are described, and to determine conformance to manufacturer's approved material fabrication, specifications and drawings. Also the inspector shall ascertain that the preparation for delivery conforms to the requirements of Section 5.

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4.5.2 Dimensional - The following components of each control unit shall be measured to ascertain that they conform to the dimensions and tolerances specified for NAEC (SI) Part No. 71-40891-4 for:

(a) Mating flange
(b) Bore and key slot of drive sleeve

4.5.3 Operation -

4.5.3.1 Full load - The motor full current draw and motor speed in each direction of rotation shall be measured when the unit is loaded by 70 foot-pound torque on a shaft inserted into the drive sleeve. The torque switch shall not be wired into the control circuit for this test. The full load current draw shall not exceed and the motor speed shall not be less than the motor nameplate ratings. The drive sleeve speed shall also be measured and shall conform to the requirements of paragraph 3.4.1.6. The drive sleeve speed may be used to compute motor speed in lieu of direct measurement.

4.5.3.2 Torque switch setting - The torque switch shall be set to open at 60 ± 6 foot-pounds when a gradually increasing load is applied to a shaft inserted into the drive sleeve. The motor control pushbutton shall be held depressed and operation of the torque switch checked to verify that the switch does not reclose prior to motor operation in the opposite direction.

4.5.3.3 Handwheel force - The handwheel force to produce 75 foot-pounds at a shaft inserted into the drive sleeve shall be measured and shall be 25 pounds maximum.

4.5.3.4 Smoothness of operation - The motor unit shall operate smoothly at full rated load and shall be free of extraordinary noises which are not consistent with or inherent in the design and normal operation of a gear train of this size and speed.

4.5.4 Shock test - Shock tests shall be conducted in accordance with MIL-S-901, Grade A, deck mounted, subsidiary component, Class I, medium weight, Type B, as specified in 4.5.4.1, 4.5.4.2, 4.5.4.3, and 4.5.4.4.

4.5.4.1 Method of mounting - Fixture 4-C as shown in MIL-S-901 shall be used to mount the test unit. The motor unit shall be mounted on the shock machine in a manner simulating a normal shipboard installation with the motor horizontal and the drive sleeve vertical.

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4.5.4.2 Mode of equipment operation during tests - The control unit shall be tested in the de-energized condition.

4.5.4.3 Definition of shock failure - The definition of failure from a shock test shall be as follows:

(a) There shall be no breakage or appreciable distortion of any parts, including the mounting bolts.

(b) If the motor unit will not deliver the required output as specified in 3.4.1.6 it shall be considered a failure and cause for rejection.

(c) Any deformation which will adversely affect operation of the unit or results in the unit functioning in any manner different than it did before the shock test shall be considered a failure and cause for rejection.

(d) The motor unit shall be disassembled following checks and inspected thoroughly for damage. Disassembly need be only to the point where the condition of the equipment can be easily observed. All observed effects of the shock test shall be carefully recorded.

4.5.4.4 Shock test sample - The motor unit which has been subjected to the shock test shall not be acceptable as a manufactured item.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging - Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A -

5.1.1.1 Cleaning - The entire motor unit shall be thoroughly cleaned in accordance with Process C-1 of MIL-P-116.

5.1.1.2 Preservation - Each motor unit shall be slushed with a preservative conforming to MIL-C-16173 Grade 3. Excess preservative shall be drained from the unit. After slushing and draining, all openings in the gear casing shall be sealed and the case partially filled with a P-2 type preservative conforming to MIL-C-16173 Grade 2.

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5.1.1.3 Unit packaging - Each motor shall be cushioned over all sharp projections with material conforming to PPP-C-843 overwrapped with Grade A Class 2 barrier material conforming to MIL-B-121 prior to being placed on the appropriate Method II container as prescribed in MIL-P-116.

5.1.2 Level C - When this level is required, each motor shall be cleaned, preserved and packaged in accordance with manufacturer's commercial practice.

5.2 Packing - Packing shall be Level A, B or C as specified. (See 6.2.)

5.2.1 Level A - The motor units preserved and packaged as specified in 5.1.1.2 and 5.1.1.3 shall be packed in overseas-type exterior shipping containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, or PPP-B-621. Plywood, when used, shall conform to Types I or II, Class 2 of NN-P-530.

5.2.2 Level B - The motor units preserved and packaged as specified in 5.1.1.2 and 5.1.1.3 shall be packed in domestic-type exterior shipping containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, or PPP-B-621.

5.2.3 Level C - Packages which require overpacking for acceptance by the carrier shall be packed in exterior-type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall conform to the Consolidated Freight Classification Rules or regulations of other common carriers, as applicable to the mode of transportation. Containers shall be able to withstand storage, rehandling and reshipment without the necessity of repacking.

5.3 Physical protection - Cushioning, blocking, bracing and bolting as required shall be in accordance with MIL-STD-1186 except that for domestic shipments, waterproofing requirements for cushioning materials and containers shall be waived.

5.4 Marking - Interior and exterior containers shall be marked in accordance with MIL-STD-129 and to include the following:

CONTROL UNIT (FOR MK-7 RETRACTABLE DECK SHEAVE)
Specification MIL-C-23771
Manufacturer's Part No.
Manufacturer's Serial No.
Name of Manufacturer

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Name of Contractor (if different from manufacturer)
Contract or Order No.
Federal Stock Catalogue Number (FSC 1710)
NAEC Part No. --

6. NOTES

6.1 Intended use - This control unit is intended to be used to actuate the retractable sheave used in the MK-7 drive system.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Quantity of control units or component parts desired
(c) Type of plans, if required by the design activity (3.4.4)
(d) Quantity of spares required (3.5)
(e) Applicable levels of packaging and packing (5.1 and 5.2)
(f) Items of data required (6.3)
(g) First article information and where the samples shall be delivered (4.3)
(h) Applicable methods of cleaning and preservation (5.1)
(i) First article sample subjected to the shock test shall not be considered or shipped as part of the contract or order (4.5)

6.2.1 Procurement requirements -

6.2.2 Contract data requirements - Items of deliverable data required by this specification are cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
3.4.4	Plans	
3.4.4.1	Approval plans	
3.4.4.2	Record plans	
4.3.1.1	First article data and disposition	
6.3.1	Manuals	
6.3.2	Provisioning documentation	
6.3.3	Plans	
6.3.4	First article inspection report	
6.3.5	Drawings and data lists	

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6.3 Data - For the information of Contractors and Contracting Officers, any of the data specified in applicable documents listed in Section 2 of this specification or referenced lower-tier documents need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order.

6.3.1 Manuals - One Instruction and Maintenance Manual shall be packaged with each control unit. The Instruction and Maintenance Manual shall be submitted for approval to the Naval Air Engineering Center, Engineering Department (SI), (Code NE-7), Philadelphia, Pennsylvania 19112 prior to acceptance of completed control units.

6.3.2 Provisioning documentation - Provisioning documentation in conformance with MIL-M-8910 (Manuals, Technical, Illustrated Parts Breakdown, Preparation of) shall be provided for inspection at the time of first article inspection. If satisfactory, the reproducible list and three copies shall be forwarded.

6.3.3 Plans - Plans as specified in 3.4.4 shall be submitted when required.

6.3.4 First article inspection report - Three copies of the first article inspection report shall be submitted to the procuring activity for approval.

6.3.5 Drawings and data list - Drawings, engineering and associated lists shall be in accordance with MIL-D-1000/1.

6.4 Marginal indicia - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Preparing activity:
Navy - AS
(DoD Project No. 1710-N007)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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